RV College of Engineering®
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Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Date	13th Jan 2026	Maximum Marks	(10+50) Marks
Course Code	IS353IA	Duration	120 Min
Sem	V Sem	Improvement CIE	
UG/PG	UG		

Artificial Intelligence and Machine Learning
(Common to CSE/ISE/CD/CY)

Q. No.	QUIZ	M	BT	CO
1.	What is the primary drawback of a standard Hill-Climbing Search algorithm, and what is the simplest modification used to mitigate it?	2	L1	CO1
2.	Briefly explain the roles of the two main operators, Crossover and Mutation, in a Genetic Algorithm's generation of a new population.	2	L2	CO1
3.	In the context of Unsupervised Cluster Evaluation, define Cohesion and Separation.	2	L1	CO2
4.	Distinguish between Hard Clustering and Soft Clustering in terms of data point assignment. Give one algorithm example for each.	2	L3	CO2
5.	The Bisecting K-means algorithm is a top-down hierarchical approach. How does it improve upon the stability and quality of results compared to a single run of standard K-means?	2	L3	CO5

Q. No.	TEST	M	BT	CO
1.	Suppose that the data mining task is to cluster the following eight points with (x, y) representing location into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9): The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. a) Write down k-means algorithm b) Apply k-means algorithm for the three cluster centers after the first round execution	10	L4	CO1
2. a	How the distance between clusters is measured in hierarchical clustering? Explain the use of this measure in making decision on when to stop the iteration	5	L2	CO1
b	Explain the primary objectives of Cluster Evaluation and the three main types of validity measures.	5	L2	CO2
3. a	What is the main dilemma when "Choosing a Cluster Validity Measure"? Outline the practical trade-offs involved.	5	L3	CO2
b	Explain two different types of clusters based on their geometry or structure, and name a suitable clustering algorithm for each.	5	L2	CO2

4. a	The goal is to maximize the value of a function f(x) where x ∈{1, 2, 3, 4, 5} and the values are f(1)=3, f(2)=7, f(3)=5, f(4)=9, f(5)=6. Show the trace of Steepest-Ascent Hill-climbing starting at x=1. Assume neighbours are x ±1.	5	L2	CO2
b	Describe the methodology involved and main operators of Genetic Algorithms .	5	L2	CO1
5. a	Explain the core principle and mechanism of Simulated Annealing. How does it guarantee completeness (eventual convergence to the global optimum)?	5	L3	CO5
b	Define Local Search Algorithms and explain why they are essential for solving Optimization Problems in large search spaces.	5	L2	CO2

Course Outcomes: After completing the course, the students will be able to: -	
CO1:	Explain and apply AI and ML algorithms to address various requirements of real-world problems
CO2:	Design and develop AI and ML solutions to benefit society, science, and industry.
CO3:	Use modern tools to create AI and ML solutions.
CO4:	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
CO5:	Conduct performance evaluation, modeling, and validation of AI and ML solutions benefiting lifelong learning